

Program : Diploma in Instrumentation Engineering	
Course Code : 5089B	Course Title: Optical Instruments Lab
Semester : 5	Credits: 1.5
Course Category: Program Elective	
Periods per week : 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- To impart practical knowledge about various properties of light.
- To demonstrate various properties of laser.
- To demonstrate different properties of optical fiber.

Course Pre-requisites:

Topic	Course code	Course name	Semester
Basic knowledge of different properties of light		Applied Physics 2	2

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Determine wave length of light/ laser, angle of prism and angle of minimum deviation using spectrometer.	16	Applying
CO2	Determine refractive index/thickness of glass slab/liquid using travelling microscope.	9	Applying
CO3	Determine wavelength of light/ radius of curvature of lens using Newton's ring apparatus.	9	Applying
CO4	Determine the numerical aperture and acceptance angle of an optical fiber.	9	Applying
	Lab Test	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	3					
CO2	3						
CO3	3						
CO4	3	3	3	3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Determine wave length of light/ laser, angle of prism and angle of minimum deviation using spectrometer.		
M1.1	Familiarize spectrometer	4	Understanding
M1.2	Determine the angle of minimum deviation of a prism	3	Applying
M1.3	Determine the angle of a prism	3	Applying
M1.4	Determine the refractive index of prism material	3	Applying
M1.5	Determine the wavelength of laser/light using spectrometer.	3	Applying
CO 2	Determine refractive index/thickness of glass slab/liquid using travelling microscope.		
M2.1	Familiarize Travelling Microscope	3	Understanding
M2.2	Determine the refractive index and thickness of glass slab.	3	Applying
M2.3	Determine the refractive index and thickness of given liquid	3	Applying
	Lab Test - I	1	
CO3	Determine wavelength of light/ radius of curvature of lens using Newton's ring apparatus.		
M3.1	Familiarize Newton's ring apparatus	3	Understanding
M3.2	Determine wavelength of light using Newton's Ring apparatus.	3	Applying

M3.3	Determine radius of curvature of lens.	3	Applying
CO4	Determine the numerical aperture and acceptance angle of an optical fiber.		
M4.1	Determine the angle of divergence of laser	3	Applying
M4.2	Determine the particle size in a powder.	3	Applying
M4.3	Determine the numerical aperture and acceptance angle of optical fibre.	3	Applying
	Lab Test - II	1	

Text / Reference:

T/R	Book Title/Author
R1	S Nagabhushana, N Sathyanarayana, Lasers and Optical Instrumentation, I K International publishing House
R2	N Subrahmanyam and Brijlal, A Textbook of Optics, S Chand Publications
R3	B L Theraja, Engineering Physics, S Chand Publications
R4	Keiser, Optical Fiber communication systems, TMH
R5	K Thyagaraja & Ghatak, Laser Theory and Applications, Springer

Online Resources:

Sl. No	Website Link
1	https://www.youtube.com/watch?v=gNOVwOthKZI
2	http://www.bsauniv.ac.in/UploadImages/Downloads/PHYSICS-LAB-
3	https://www.youtube.com/watch?v=hK7sWjaFT1M
4	https://www.learnbse.in/to-determine-refractive-index-of-a-glass-slab-using-a-
5	http://www.iiserpune.ac.in/~bhasbapat/phy221_files/Gratings%20and%20Prism%20Spectrometer.pdf